

Topic2-1

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Multiple Linear Regression



Photo by **Erik Eastman** on **Unsplash**

Last week

SLR: definition, estimation and inference

- We view examples where a linear regression can be used to study the relationship between a response variable and input variables of different types
- We discussed and identified different uses of linear regression models (estimation vs inference vs prediction)
- We defined a simple linear regression (SLR, a linear regression with only one input variable)
- We defined, interpreted and visualized the different terms of a linear regression
- We learned that least squares is a method that can be used to estimate a linear regression using data
- We learned two different ways to make inference in the context of SLR:
 - using theoretical results or using bootstrapping
- We used computer scripts for **estimation** and **inference** tasks in a SLR analysis

This week

Multiple Linear Regression (MLR)

This week we are going to "unleash" the scope of linear regression models and use them to study the association between a continuous response and *many* input variables of *different types*!!

A linear regression model with many input variables is usually called a **Multiple Linear Regression (MLR)**.

Heads up: *Multiple Linear Regression is not the same as Multivariate Linear Regression. The latter refers to a regression with a multivariate response variable (beyond the scope of this course)!!*

- 1. Categorical input variables with 2 or more levels**
- 2. Additive MLR: with different type of input variables**
- 3. MLR with interaction terms: interactions between continuous and categorical input variables**

Learning Objectives

By the end of this lecture you will be able to:

1. Give an example of a real problem that that could be answered by a multiple linear regression.
2. Interpret the coefficients and p -values of different types of input variables, including categorical input variables.
3. Define interactions in the context of linear regression.
4. Write a computer script to perform linear regression when input variables are continuous or discrete, and when there are interactions between some of these variables.

1. Categorical input variables *predictors*

In previous courses you've learned that a categorical variables creates groups (levels) and you can study how another variable differ among these groups.

for example: does the size of the donations depend on the variation of a website?

You can extend this notion to LR, is the linear relation the same among groups?

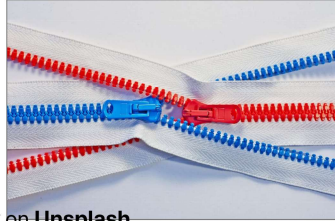


Photo by Tomas Sobek on Unsplash

Let's use the `US_cancer_data` to explore if the cancer mortality differ by states.

Note: the original `US_county_cancer_data` does not contain a specific column for the state where each county belongs, so we'll get that information from the variable `Geography` using code to split strings.

In []:

```
US_cancer_data$state <- sapply(strsplit(US_cancer_data$Geography, ", "), "[", 2)
```

In [4]:

```
head(US_cancer_data, 3)
```

A tibble: 3 × 5

TARGET_deathRate	povertyPercent	PctPrivateCoverage	Geography	state
<dbl>	<dbl>	<dbl>	<chr>	<chr>
164.9	11.2	75.1	Kitsap County, Washington	Washington
161.3	18.6	70.2	Kittitas County, Washington	Washington
174.7	14.6	63.7	Klickitat County, Washington	Washington

1.1. One categorical variable with 2 levels ← as predictor

Let's start by comparing the mortality rates in 2 states: Washington vs Indiana.

state = $\begin{matrix} 0 \\ 1 \end{matrix}$ $\begin{matrix} \text{Indiana} \\ \text{Washington} \end{matrix}$

In [5]:

```
WI_cancer_data <- US_cancer_data %>% filter(state %in% c("Washington", "Indiana")) %>%  
  droplevels()  
WI_cancer_data$state <- as.factor(WI_cancer_data$state)
```

categorical variable

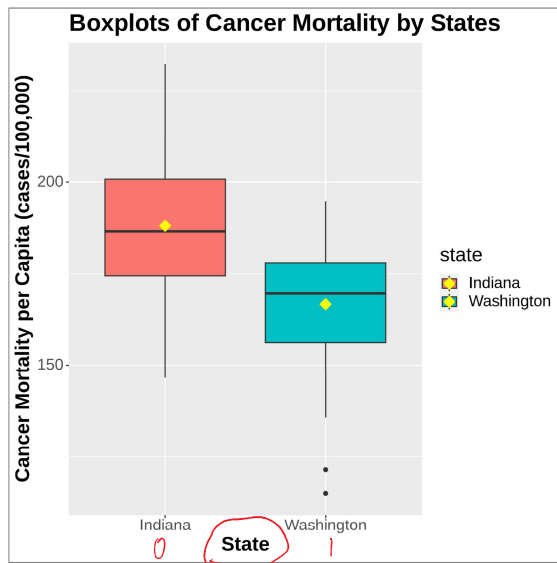
Visualization

In []:

```
TARGET_deathRate_boxplots <- WI_cancer_data %>%
  ggplot() +
  geom_boxplot(aes(state, TARGET_deathRate, fill = state)) +
  theme(
    text = element_text(size = 18),
    plot.title = element_text(face = "bold"),
    axis.title = element_text(face = "bold")
  ) +
  ggtitle("Boxplots of Cancer Mortality by States") +
  xlab("State") +
  ylab("Cancer Mortality per Capita (cases/100,000)") +
  stat_summary(aes(state, TARGET_deathRate, fill = state),
    fun = mean, colour = "yellow", geom = "point",
    shape = 18, size = 5
  )
```

In [7]:

```
TARGET_deathRate_boxplots
```



$y = \beta_0 + \beta_1 \text{state} + \varepsilon$
t-test
equivalent

The side-by-side boxplots in `TARGET_deathRate_boxplots` show some difference in mortality rates between these two states.

Can we use a linear regression to study the variation of mortality between different states??!!

The linear regression equation is:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

here, the response Y_i is `TARGET_deathRate` and we use i to identify the i th state

The regression coefficients β_0 and β_1 are the **true and unknown** coefficients we want to estimate.

But the x -axis in `TARGET_deathRate_boxplots` is not numeric: Indiana vs Washington

There's not a *line* in this case!! However, we can still call use a "linear regression"!!

How??

Trick: we will use an auxiliary numeric variable to represent the levels of a categorical variable: a dummy variable

The function `lm` creates this variable for you if you indicate that the input variable (in our case `state`) is a factor!

Dummy variables

What is a dummy variable? It is a *numerical variable* that could either take on the values 0 or 1. For this specific example, for the i th state, X_i can be defined as follows:

$$\text{stateWashington}_i = X_i = \begin{cases} 1 & \text{if the county is in "Washington";} \\ 0 & \text{if otherwise} \end{cases}$$

Heads-up: The level in the dummy variable corresponding to the value of 0 is called the reference (or baseline) level

CODE

If `state` is a factor, `lm` creates these dummy variables for you!

- `lm` calls the dummy variable `stateWashington` (name of the variable followed by the level corresponding to 1)
- The reference level (dummy variable = 0, level "left out") is "Indiana", chosen alphabetically.

In []:

```
WI_data_LR <- tidy(lm(TARGET_deathRate ~ state,  
  data = WI_cancer_data)) %>% mutate_if(is.numeric, round, 3)
```

In [9]:

WI_data_LR

A tibble: 2 × 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	188.121	1.825	103.074	0
stateWashington	-21.628	3.345	-6.466	0

The inclusion of the dummy variable creates 2 groups

- For counties in Indiana, $X_i = 0$, then $Y_i = \beta_0 + \beta_1 \times 0 + \varepsilon_i$

$$Y_i = \beta_0 + \varepsilon_i$$

- For counties in Washington, $X_i = 1$, then $Y_i = \beta_0 + \beta_1 \times 1 + \varepsilon_i$

$$Y_i = \beta_0 + \beta_1 + \varepsilon_i$$

$$y = \beta_0 + \beta_1 x + e$$

$$x = \begin{cases} 0 & \text{Indiana} \\ 1 & \text{WA} \end{cases}$$

Interpretation:

A bit of math to understand the meaning of the coefficients

- For counties in Indiana, $X_i = 0$,

$$E[Y_i | X_i = 0] = \beta_0$$

Important: β_0 is the **mean of the response for the reference level** of the input variable.

for example, the mean mortality rate in Indiana

- For counties in Washington, $X_i = 1$,

$$E[Y_i | X_i = 1] = \beta_0 + \beta_1$$

Note that

$$\beta_1 = E[Y_i | X_i = 1] - E[Y_i | X_i = 0]$$

Important: β_1 is the ***difference* of means of the response between levels**.

for example, the difference between the mean mortality rate in Washington with respect to that in Indiana

Note that β_0 and β_1 in this particular case are *not* intercepts and slopes (even if R calls them that way)

random → fixed → random

$$y = \beta_0 + \beta_1 x + e$$

$E(y | x)$

$$= \beta_0 + \beta_1 x$$

equivalent fixed, not random

In [10]:

```
WI_data_LR
```

A tibble: 2 × 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	188.121	1.825	103.074	0
stateWashington	-21.628	3.345	-6.466	0

INTERPRETATION OF THE ESTIMATED COEFFICIENTS

- The estimated $\hat{\beta}_0 = 188.121$, is the average (sample mean) mortality rate rate per capita in Indiana!!

It is the sample version of the conditional expectation (mean of the reference group)

- The estimated $\hat{\beta}_1 = -21.628$ is the difference between the average mortality rate per capita in Washington and the average mortality rate per capita in Indiana.

It is the sample version of the difference of the conditional expectations (or group means)

$$\beta_1 = E(y|x=1) - E(y|x=0)$$

NOTE: THAT YOU HAVE DONE THESE TYPE OF ANALYSES IN STAT201!!

In [11]:

```
t.test(TARGET_deathRate ~ state,WI_cancer_data,var.equal=T)
```

Two Sample t-test

```
data: TARGET_deathRate by state
t = 6.4659, df = 129, p-value = 1.896e-09
alternative hypothesis: true difference in means between group Indiana and
group Washington is not equal to 0
95 percent confidence interval:
 15.01026 28.24643
sample estimates:
mean in group Indiana mean in group Washington
      188.1207           166.4923
```

This is not a coincidence! $\ln$ is running the *same* t -test!!

1.2 One categorical variable with more than 2 levels

But what do we do if the categorical variable has more levels??!

We need additional dummy variables! Dummy variables are comparisons to a *reference* level. Thus, we need more of these variables to compare all other levels with respect to the reference.

Suppose we want to compare "Indiana", "Washington" and "Kansas":

$$\text{stateWashington}_i = X_{1i} = \begin{cases} 1 & \text{if the county is in "Washington";} \\ 0 & \text{if otherwise} \end{cases}$$

$$\text{stateKansas}_i = X_{2i} = \begin{cases} 1 & \text{if the county is in "Kansas";} \\ 0 & \text{if otherwise} \end{cases}$$

$X = \begin{cases} 0 & \text{Indiana} \\ 1 & \text{Washington} \\ 2 & \text{Kansas} \end{cases}$

X_1, X_2
dummy variables

Indiana
↓
reference category is hidden

$$y = \beta_0 + \beta_1 X + e \quad \Longleftrightarrow \quad y = \beta_0 + \beta_1^* X_1 + \beta_2^* X_2 + e$$

$(= 1, 2, \dots, n)$
equivalent

Note that we need two dummy variable for 3 levels (2 levels compared to one reference level)

- The reference level is the same for both dummy variables and equal to 0.
- The other levels equal 1 in each dummy variable.

↓ $x = 0$, (Indiana)

In [12]:

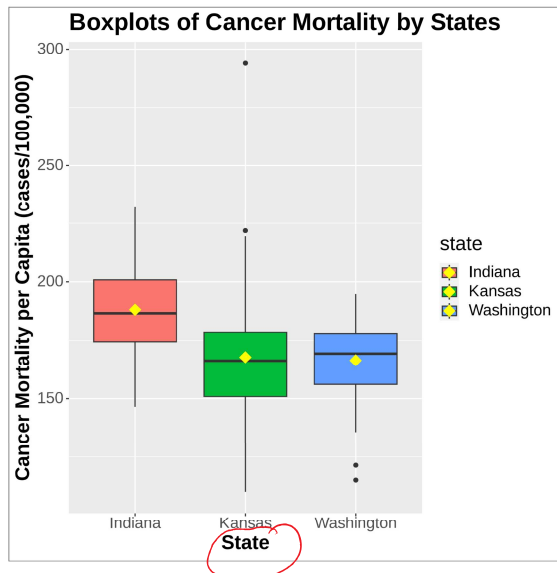
```
WIK_cancer_data <- US_cancer_data %>%filter(state %in% c("Indiana", "Washington", "Kansas"))%  
  droplevels()  
WIK_cancer_data$state <- as.factor(WIK_cancer_data$state)  
str(WIK_cancer_data)
```

```
tibble [233 × 5] (S3: tbl_df/tbl/data.frame)  
$ TARGET_deathRate : num [1:233] 165 161 175 195 144 ...  
$ povertyPercent   : num [1:233] 11.2 18.6 14.6 17.1 12.5 15.6 23.2 17.8  
22.3 13.1 ...  
$ PctPrivateCoverage: num [1:233] 75.1 70.2 63.7 58.4 61.6 60 49.5 55.8 5  
5.5 69.9 ...  
$ Geography        : chr [1:233] "Kitsap County, Washington" "Kittitas Co  
unty, Washington" "Klickitat County, Washington" "Lewis County, Washington"  
...  
$ state             : Factor w/ 3 levels "Indiana","Kansas",...: 3 3 3 3 3  
3 3 3 3 3 ...
```

Visualization

In [16]:

```
TARGET_deathRate_boxplots
```



ANOVA

The inclusion of the 2 dummy variable creates 3 groups

- For counties in Indiana, $X_i = 0$, then $Y_i = \beta_0 + \beta_1 \times 0 + \beta_2 \times 0 + \varepsilon_i$

$$Y_i = \beta_0 + \varepsilon_i$$

- For counties in Washington, $X_i = 1$, then $Y_i = \beta_0 + \beta_1 \times 1 + \beta_2 \times 0 + \varepsilon_i$

$$Y_i = \beta_0 + \beta_1 + \varepsilon_i$$

- For counties in Kansas, $X_i = 1$, then $Y_i = \beta_0 + \beta_1 \times 0 + \beta_2 \times 1 + \varepsilon_i$

$$Y_i = \beta_0 + \beta_2 + \varepsilon_i$$

$$X_i = \begin{cases} 0 \\ 1 \end{cases}$$

$$X_i = \begin{cases} 0 \\ 1 \end{cases}$$

$$X_i$$

Interpretation:

- β_0 is the mean of the response for the *reference* level of the input variable. In our case, the mean mortality rate in Indiana
- β_1 is the difference between the mean mortality rate in Washington, and that in Indiana
- β_2 is the difference between the mean mortality rate in Kansas and that in Indiana

Note: you can check numerically that the **estimates** of these parameters are based on the sample means!!

Interpretation
in words

$$\beta_2 = E(y|x=2) - E(y|x=0)$$

C.I.: estimate $\pm$ 1.96 SE,

Testing: $Z = \frac{\text{estimate}}{SE}$

In [13]:

```
WIK_data_LR <- tidy(lm(TARGET_deathRate ~ state,  
  data = WIK_cancer_data)) %>% mutate_if(is.numeric, round, 3)
```

In [17]:

WIK_data_LR

A tibble: 3 x 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	188.121	2.291	82.127	0
stateKansas	-20.286	3.159	-6.422	0
stateWashington	-21.628	4.198	-5.152	0

X_1

X_2

estimate

SE

accuracy of estimate

most important

$\rightarrow \sim N(0,1)$

$y = \beta_0 + \beta_1 X + e$

$X \Rightarrow X_1, X_2$

2. MLR: additive models

Last week we studied the relation between mortality, `TARGET_deathRate`, percentage of the county's populace in poverty (`povertyPercent`) is positively association.

Does this relation change among states??

Should we study these relations separately or jointly using linear regressions?

Categorical variables create groups to study these linear relations!

2.1 One continuous and one categorical input variables

There are different ways of adding variables in a LR: with or without interaction

When a variable is just added (without interaction), we assume that its association with the response **does not** depend on other variables

For example, the expected change in mortality per percentage change in populace poverty is the same in all states

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + e$$

=  additive model.

Let's continue with the example of the 2 states

$$Y_i = \beta_0 + \beta_1 \text{stateWashington}_i + \beta_2 \text{povertyPercent}_i + \varepsilon_i,$$

Again, the dummy variable creates 2 groups and in this case 2 lines:

- For counties in Indiana, $Y_i = \beta_0 + \beta_1 \times 0 + \beta_2 \times \text{povertyPercent}_i + \varepsilon_i$

$$Y_i = \beta_0 + \beta_2 \times \text{povertyPercent}_i + \varepsilon_i$$

Handwritten notes:

$y = \beta_0 + \beta_1 \text{state} + \beta_2 \text{poverty} + \textcircled{e}$

Annotations: An arrow points from the word "uncertainty, noise" to the circled e . Another arrow points from the word "state" to β_1 . A third arrow points from the word "poverty" to β_2 . There is an equals sign (=) written below the y .

- For counties in Washington, $Y_i = \beta_0 + \beta_1 \times 1 + \beta_2 \times \text{povertyPercent}_i + \varepsilon_i$

$$Y_i = (\beta_0 + \beta_1) + \beta_2 \times \text{povertyPercent}_i + \varepsilon_i$$

Note: that both lines have the same slope β_2 !!

We assumed that the expected change in mortality per percentage change in populace poverty is the same in both states

INTERPRETATION:

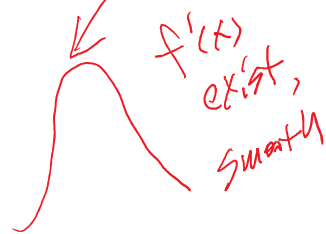
- β_0 is intercept of the *reference* line.
- β_1 (coefficient of the dummy variable) is the difference between intercepts of both lines
- β_2 is the *common* slope of both lines

First-order

✓ Taylor approximation:

For any smooth function,

$$f(x) \approx \underbrace{x_0 + f(x_0)x}_{\text{linear}} + \text{remaining terms}$$



$$y = f(x_1^*, x_2) + e.$$

only

In [20]:

```
MLR_state_poverty_add <- tidy(lm(TARGET_deathRate ~ state + povertyPercent,
  data = WI_cancer_data)) %>% mutate_if(is.numeric, round, 3)
```

In [21]:

```
MLR_state_poverty_add
```

A tibble: 3 × 5

term	estimate	std.error	statistic	p.value
<chr>	<dbl>	<dbl>	<dbl>	<dbl>
(Intercept)	170.199	5.655	30.096	0.000
stateWashington	-24.542	3.337	-7.353	0.000
povertyPercent	1.300	0.390	3.334	0.001

$$y = \beta_0 + \beta_1 \text{state} + \beta_2 \text{poverty} + e$$

slope

NOTE: THERE ARE 3 COEFFICIENTS FOR 2 LINES BECAUSE THE ADDITIVE MODEL ASSUMES A COMMON SLOPE!!

Non-additive model:

$$y = \beta_0 + \beta_1 \left(\frac{x_1}{x_2} \right) + e$$

linear

2.2 Adding continuous variables

We can also add continuous variables in a MLR. For example, the mortality rate may also depend on the percentage of the population with private health coverage.

$$Y_i = \beta_0 + \beta_1 \text{povertyPercent}_i + \beta_2 \text{PctPrivateCoverage}_i + \varepsilon_i$$

Important: we are again assuming that the expected change in the response per unit change in an input does not depend on the value of other variables!!

$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e$
Interpretation: of β_1 : ^{average values of x_1, x_2 continuous} change in y when x_1 changes by one unit, assuming x_2 is fixed.
(or adjusted for x_2 , or incorporating x_2 info)

For example:

If we hold the poverty percentage, say at 15:

- the relationship between mortality and the percentage with private coverage is
linear

$$Y_i = \beta_0 + \beta_1 15 + \beta_2 \text{PctPrivateCoverage}_i + \varepsilon_i$$

- the slope β_2 *does not depend* on the value held constant

(and same for holding PctPrivateCoverage fix)

THESE MODELS ARE CALLED *ADDITIVE*. THINK ABOUT THIS ASSUMPTIONS WHEN YOU SOLVE EXERCISES IN THE WORKSHEET!

What have we learned today?

- LR can include categorical variables
- Numerical variables (dummy variables) are created to represent the levels of categorical variables
- Categorical variables create groups to analyze the linear associations within groups
- Additive models with a categorical variable assume common slopes i.e., the expected change in the response per unit change in the input variable does not depend on the level of the categorical variable(s)
- Additive models assume (in general) that the expected change in the response per unit change in an input does not depend on the value of other variables!!

categories

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e$$